

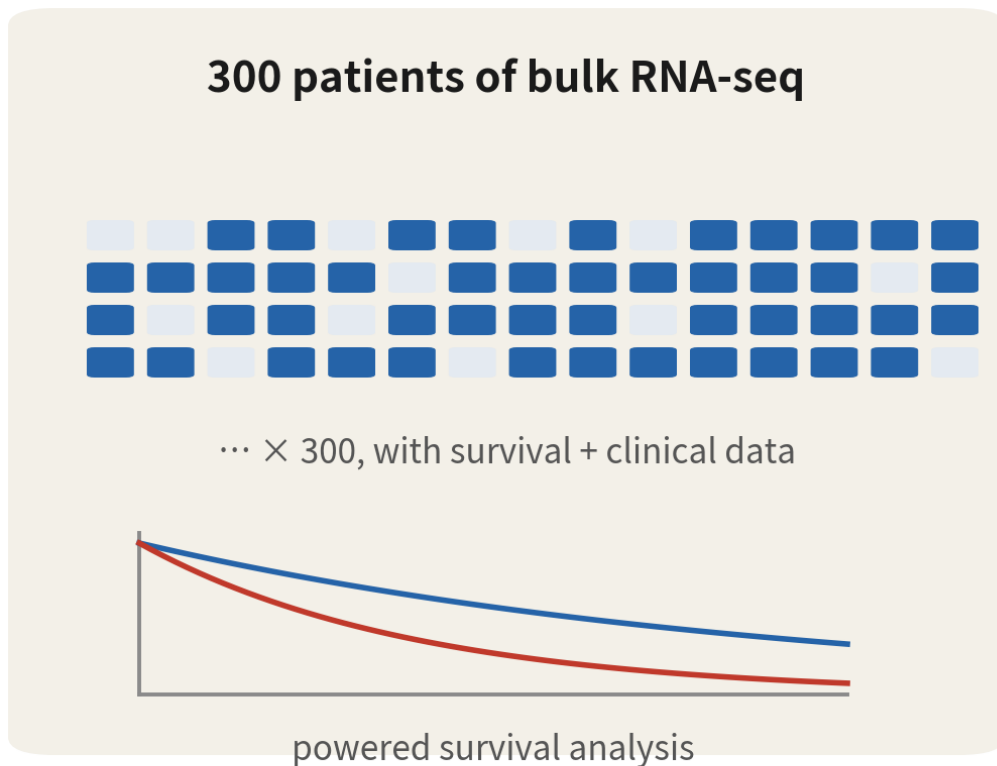
B17 Series B

Deconvolution: unmixing bulk data with a single- cell reference

Build your own ground truth and grade your own answers — measure the error, then choose a method.

About 38 min | Prerequisites: B10, B12 | Data: pbmc3k pseudo-bulks

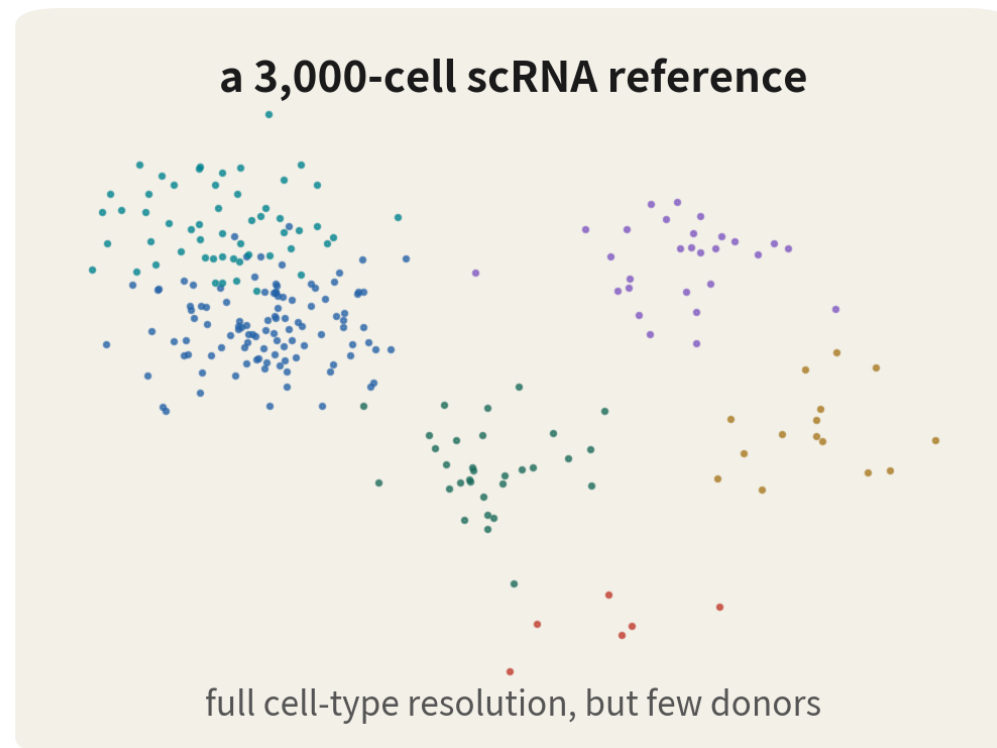
300 patients vs 3,000 cells



who is who
(signatures)



cohort size and
clinical endpoints



deconvolution: let each side complete the other

Bulk has the cohort and the endpoints; single cell has the resolution. Why not let each complete the other?

WHERE THIS EPISODE STARTS

**Every number in a bulk profile
is the sum of a mixture.**

Deconvolution splits that soup back into cell-type shares — and you happen to hold the key.

What you will learn

- 1 State the mixture equation and how it mirrors B12's pseudobulk
- 2 Build a signature and 20 known-proportion pseudo-bulks from pbmc3k, run NNLS, and grade it
- 3 Place CIBERSORTx / MuSiC / Bisque / SPOTlight, and predict the bias when the reference misses a type

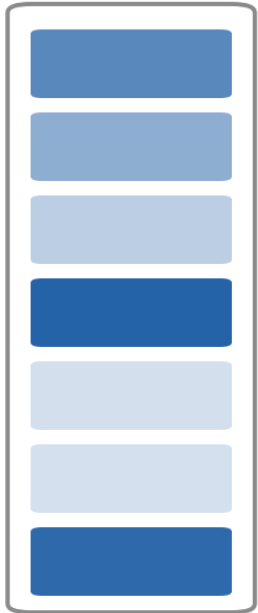
01

One equation, two directions

bulk $\approx$ signature $\times$ proportions — every tool is a variation on it

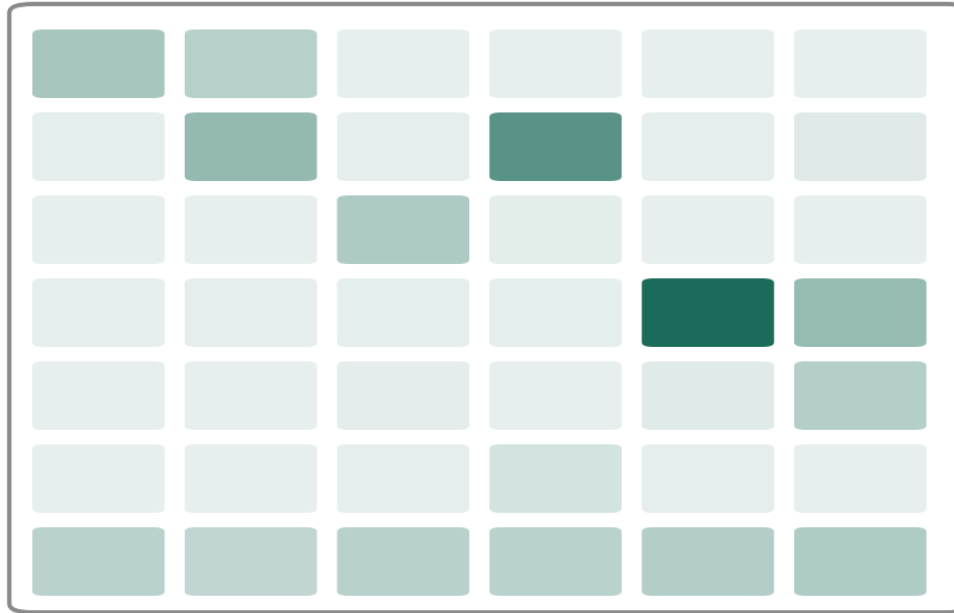
The mixture equation

bulk expression



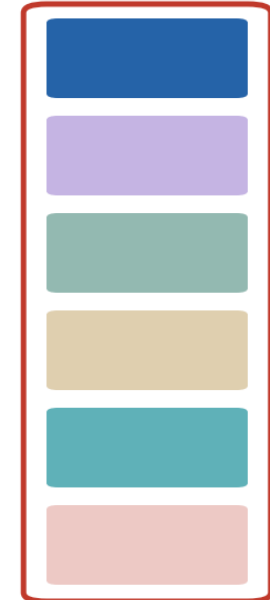
genes $\times$ 1 (measured)

signature matrix S



genes $\times$ types (from scRNA)

proportions p



CD4 T

CD8 T

B

NK

Mono

DC

types $\times$ 1 (the unknown)

$\approx$

$\times$

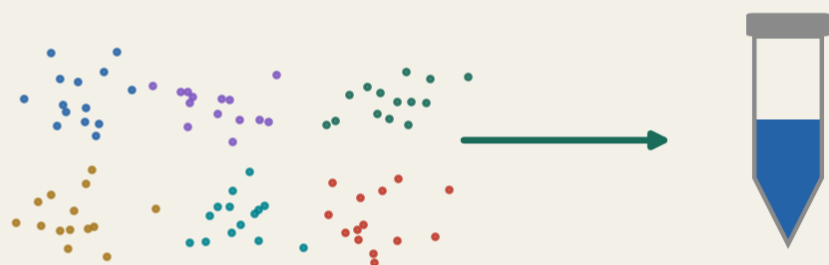
deconvolution = given bulk and S , solve for p

Given the bulk vector and the signature matrix S , solve for the proportions p — that is the whole episode

B12 sums up, B17 splits back

B12: pseudobulk (summing up)

each cell's label is known

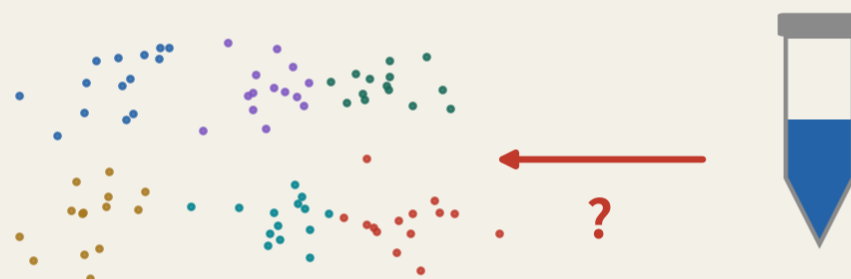


known: cells $\times$ labels

compute: the bulk vector

B17: deconvolution (splitting back)

only the mixed bulk; labels are gone



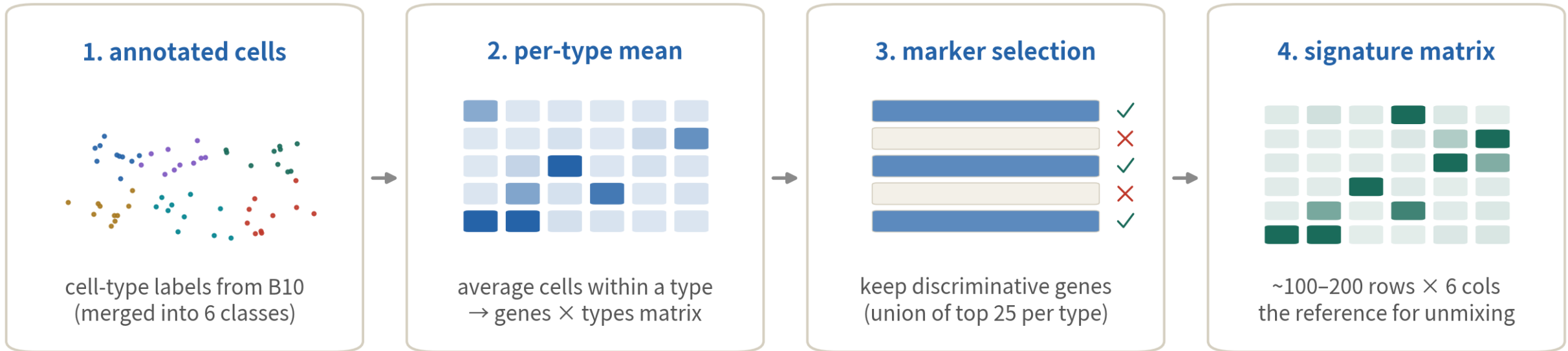
known: bulk vector + S

solve: proportions p

one equation: $\text{bulk} \approx S \times p$ — B12 verifies it, B17 solves it

Pseudobulk sums labelled cells into bulk; deconvolution splits unlabelled bulk into shares — the same identity

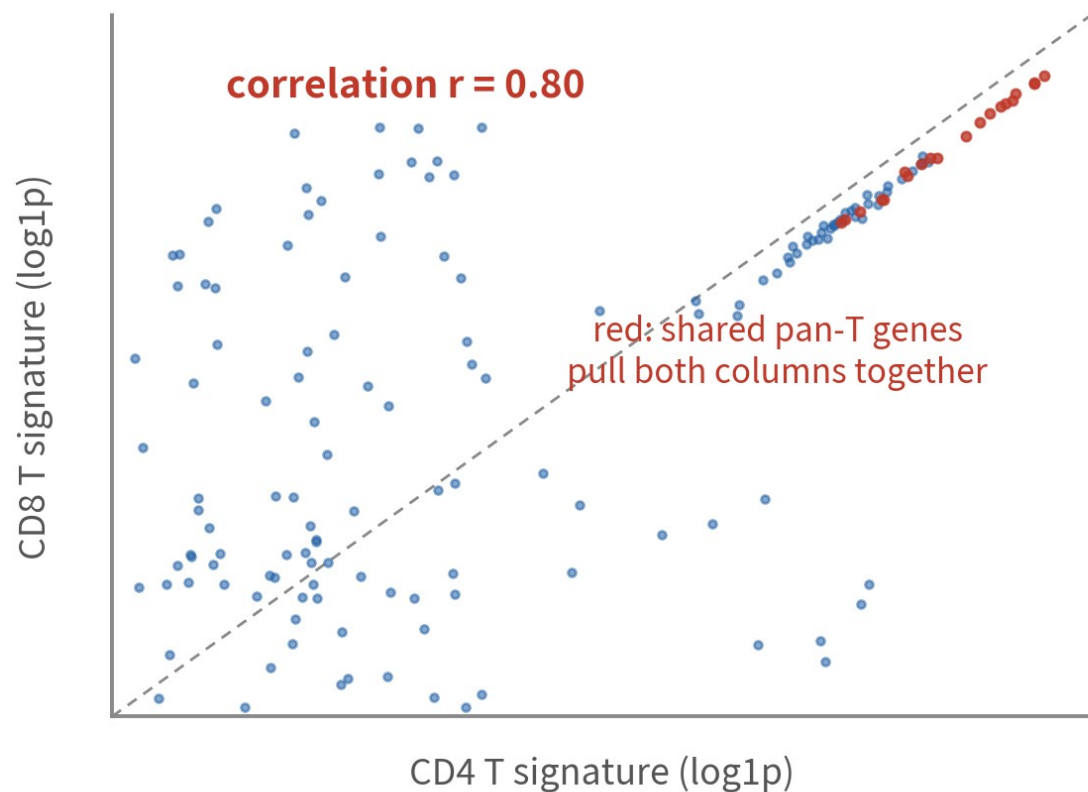
Where the signature matrix comes from



marker selection is not cosmetic — undiscriminative genes make the system harder to solve

- B10's labels → per-type mean expression
- marker selection: keep only discriminative genes
- housekeeping rows are uniformly bright — zero signal, drop them
- resolution is set by the signature, not by the tool

Collinearity: CD4 T and CD8 T look alike



column-column correlation

CD4 T	1.00	0.80	-0.05	-0.07	-0.02	0.03
CD8 T	0.80	1.00	-0.16	0.36	-0.15	-0.12
B	-0.05	-0.16	1.00	-0.07	-0.02	0.03
NK	-0.07	0.36	-0.07	1.00	-0.05	-0.00
Mono	-0.02	-0.15	-0.02	-0.05	1.00	0.57
DC	0.03	-0.12	0.03	-0.00	0.57	1.00
	CD4 T	CD8 T	B	NK	Mono	DC

simulated data

Shared pan-T genes push the two columns to $r = 0.8$ — the system turns ill-conditioned; the two shares can trade places at little cost

Two kinds of output — do not conflate them

First question for any result: which kind is this?

absolute

estimates a share of all cells

- what the NNLS / MuSiC route aims for
- sums to 1; comparable across types
- still distorted by RNA content differences

relative score

compare one type across samples

- CIBERSORT's default output is of this kind
- rank samples, but never compare across types
- a chronic source of misreading in papers

CORE IDEA

**The ceiling of deconvolution is the signature,
not the algorithm.**

A mismatched or incomplete or collinear reference cannot be rescued by any solver. So we build
S with care first.

02

Build the signature from pbmc3k

B10 labels → merge classes → average → pick markers

Load the annotated object, merge to 6 classes

```
library(Seurat)
library(dplyr)
set.seed(1234)

pbm.c <- readRDS("output/pbm.c_b10_annotated.rds")
# 亞群太細拆不動：先合併成 6 大類（理由見下一頁）
merge.map <- c("Naïve CD4 T" = "CD4 T", "Memory CD4 T" = "CD4 T",
              "CD8 T" = "CD8 T", "Effector CD8 T" = "CD8 T",
              "CD14 Mono" = "Mono", "FCGR3A Mono" = "Mono",
              "B" = "B", "NK" = "NK", "DC" = "DC")
pbm.c$ct <- merge.map[as.character(pbm.c$celltype.manual)]
pbm.c <- subset(pbm.c, subset = !is.na(ct)) # Platelet 太少，退場
table(pbm.c$ct)
```

B	CD4 T	CD8 T	DC	Mono	NK
344	1178	311	32	594	165

The first decision is not which tool — it is how fine to split

Per-type mean × marker filter → signature

```
Idents(pbm c) <- "ct"
m k <- FindAllMarkers(pbm c, only.pos = TRUE,
                      min.pct = 0.25, logfc.threshold = 1)
top <- m k %>% group_by(cluster) %>%
  slice_max(avg_log2FC, n = 25) %>% pull(gene) %>% unique()

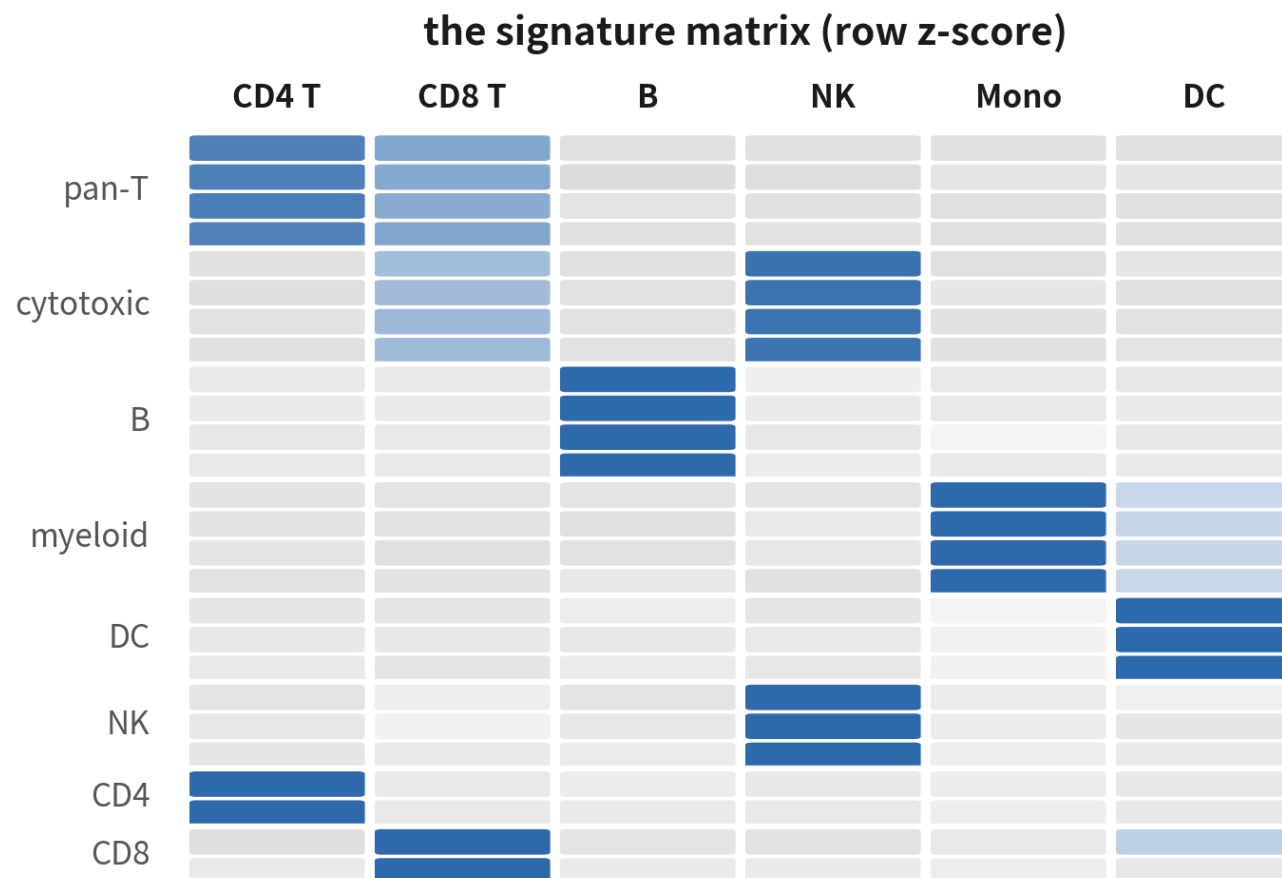
avg <- AggregateExpression(pbm c, group.by = "ct")$RNA # 加總 counts
sig <- t(t(avg) / colSums(avg)) * 1e4 # 每欄 CPM 化
sig <- sig[intersect(top, rownames(avg)), ]
dim(sig); round(cor(sig)["CD4 T", "CD8 T"], 2)
```

```
[1] 138 6
```

```
[1] 0.8
```

Check column correlations first: CD4 vs CD8 is already 0.8 — this collinearity tax will show up soon

The finished signature, visualised



diagonal blocks

each type owns a bright block
→ the system is solvable

shared blocks

pan-T and cytotoxic modules span columns
→ CD4/CD8 and CD8/NK entangle

housekeeping

uniform rows carry no signal
→ drop them from the signature

simulated data

Diagonal blocks make the system solvable; the pan-T and cytotoxic cross-column blocks are where collinearity lives

What this section did

6 cell types



B10's 10 labels merged
into 6 solvable classes

one signature



138 markers $\times$ 6
columns, CPM per
column

one warning



$\text{cor}(\text{CD4}, \text{CD8}) = 0.8$ —
collinearity on record

03

Manufacture your own ground truth

sample cells, mix 20 pseudo-bulks at known proportions, save the answer key

Mix 20 pseudo-bulks at known proportions

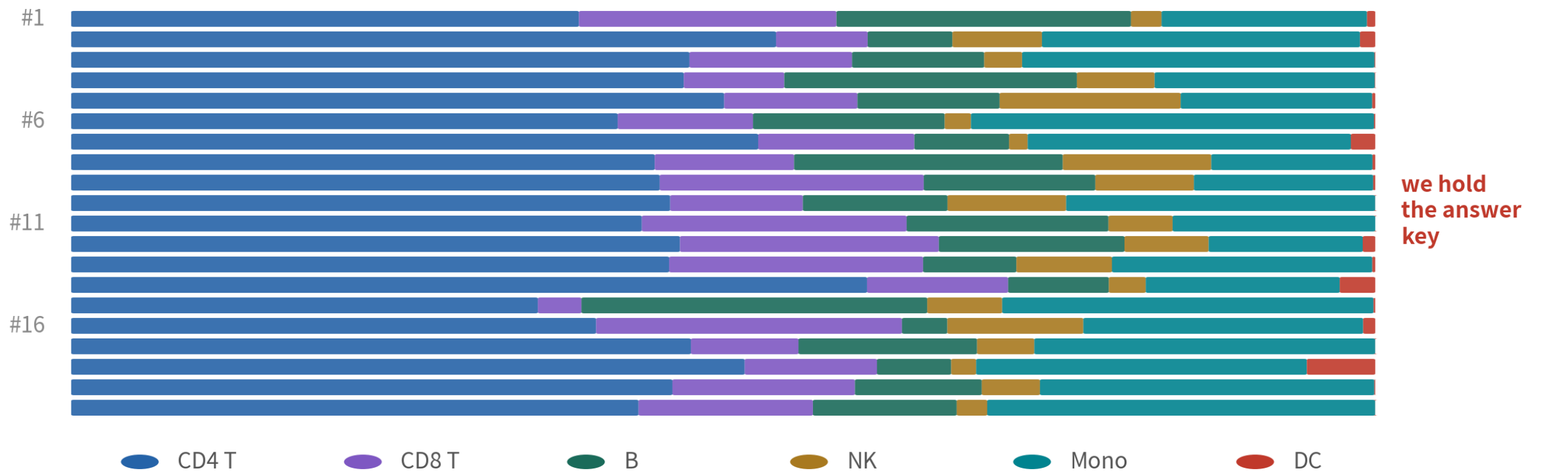
```
types <- colnames(sig)
alpha <- as.numeric(table(pbm.c$ct)[types]) / ncol(pbm.c) * 80
truth <- t(replicate(20, { g <- rgamma(6, alpha); g / sum(g) }))
colnames(truth) <- types # 真值表: 對答案用

cnt <- GetAssayData(pbm.c, layer = "counts")
bulk <- sapply(1:20, function(b) {
  cells <- unlist(lapply(types, function(tp)
    sample(colnames(pbm.c)[pbm.c$ct == tp],
      round(500 * truth[b, tp]), replace = TRUE)))
  Matrix::rowSums(cnt[, cells]) # 500 顆加總 = 一個假 bulk
})
bulk <- t(t(bulk) / colSums(bulk)) * 1e4 # 每個 bulk 也 CPM 化
```

Dirichlet draws make the 20 bulks differ; the proportions are saved the moment they are drawn — we hold the key

The 20 bulks' ground truth

the 20 pseudo-bulks' TRUE proportions (Dirichlet draws, saved to disk)



simulated data

Each row is one bulk's true composition — NNLS will be graded against exactly this table

What you hold at this point

a bulk matrix



genes $\times$ 20 — each
column is a mixed
soup

the answer key



20 $\times$ 6 proportions,
each row sums to 1

nothing solved yet



now pretend you
never saw the key, and
solve for p

04

NNLS: the core solver in under ten lines

transparent, explainable, accurate enough — understand it before upgrading

What NNLS does

ordinary least squares

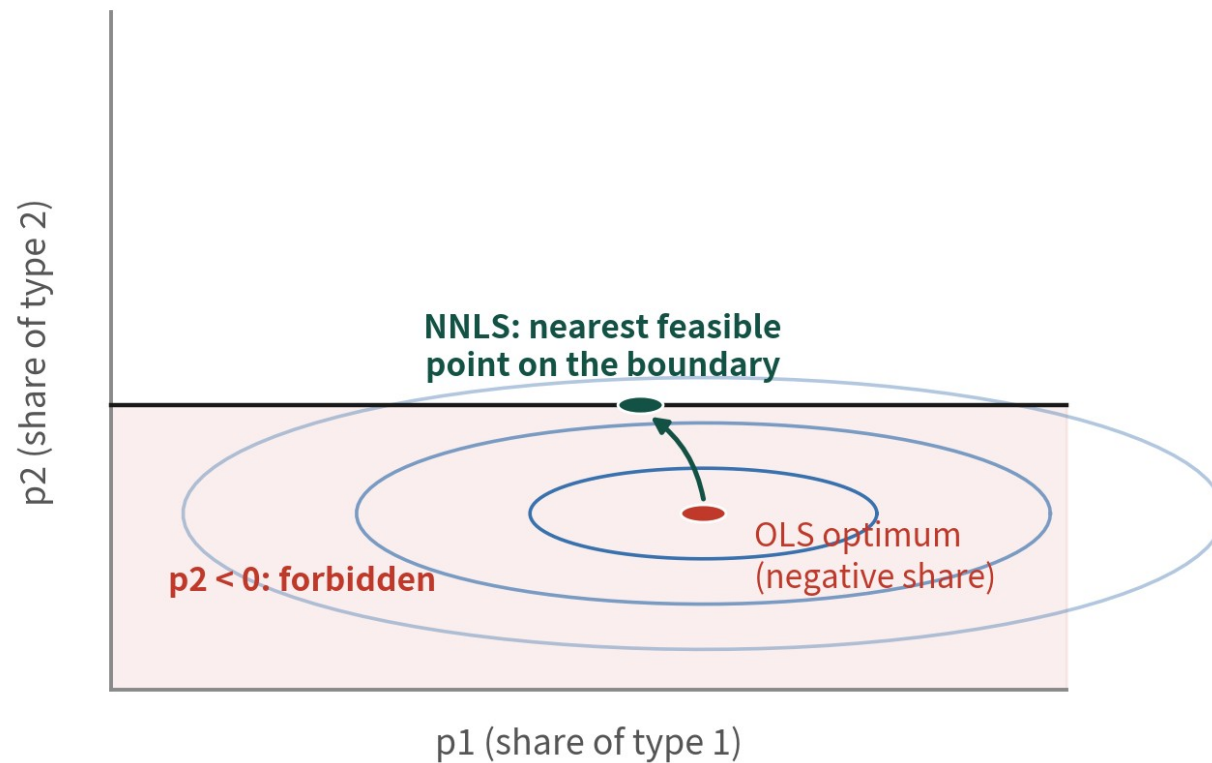
minimise $\| \text{bulk} - S \times p \|$

but p may go negative — '-8% NK' is meaningless

non-negative least squares

same objective, one constraint: $p \geq 0$

normalise the solution to sum 1 $\rightarrow$ proportions



Same objective as ordinary least squares plus one constraint — no negative shares; normalise the solution to sum 1

The core solve: one bulk

```
# install.packages("nnls")
library(nnls)

A <- sig          # 基因 × 6 型別
b <- bulk[row names(sig), 1] # 第 1 個假 bulk，取同一組基因

fit <- nnls(A, b)      # 非負最小平方：核心就這一行
prop <- fit$x / sum(fit$x) # 正規化成總和 = 1
round(setNames(prop, types), 3)
```

```
CD4 T CD8 T   B   NK Mono   DC
0.388 0.196 0.219 0.030 0.154 0.012
```

This bulk's truth is 0.389 / 0.198 / 0.226 / 0.023 / 0.158 / 0.006 — feel the distance

Loop over all 20 bulks

```
est <- t(apply(bulk[row.names(sig), ], 2, function(b) {  
  x <- nnls(A, b)$x  
  x / sum(x)  
}))  
colnames(est) <- types  
head(round(est, 3), 3)
```

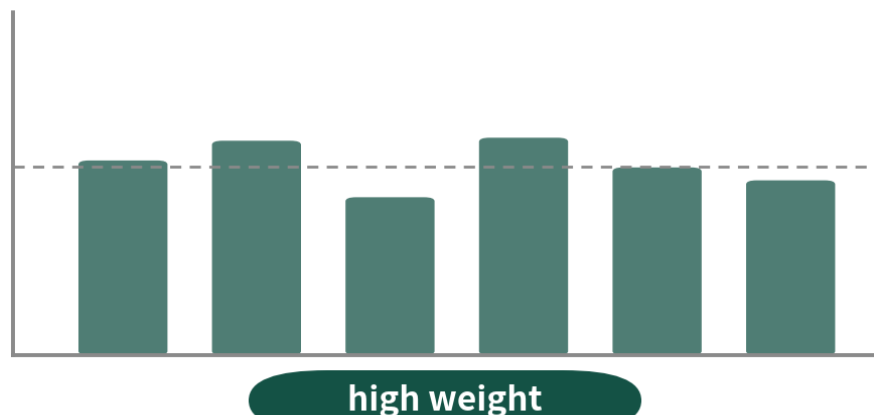
```
   CD4 T CD8 T   B   NK Mono   DC  
[1,] 0.388 0.196 0.219 0.030 0.154 0.012  
[2,] 0.554 0.053 0.070 0.072 0.230 0.022  
[3,] 0.509 0.077 0.106 0.053 0.246 0.008
```

Twenty bulks at milliseconds each — deconvolution was never a compute problem

What MuSiC adds: weighting

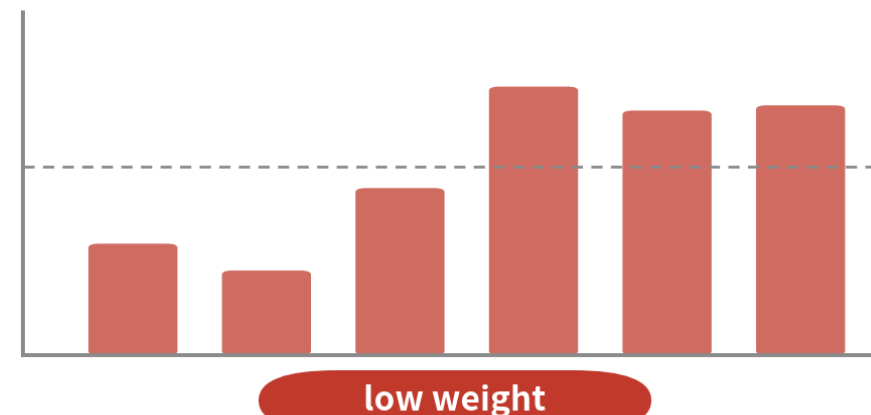
gene A: stable across subjects

one marker $\times$ 6 donors



gene B: noisy across subjects

one marker $\times$ 6 donors



MuSiC keeps all genes but weights them continuously —

genes that are stable across subjects and informative speak louder in the fit

simulated data

Stable-across-subjects genes speak louder, noisy ones quieter — same NNLS skeleton, redistributed votes

05

Grade the answers

estimate vs truth — overall first, then per type

The overall report card

```
library(tidyrr)
df <- bind_rows(
  as.data.frame(truth) |> mutate(bulk = 1:20, what = "truth"),
  as.data.frame(est) |> mutate(bulk = 1:20, what = "est")) |>
  pivot_longer(all_of(types), names_to = "ct") |>
  pivot_wider(names_from = what, values_from = value)

cor(df$truth, df$est)          # 整體相關
sqrt(mean((df$truth - df$est)^2)) # RMSE : 平均差幾個百分點
```

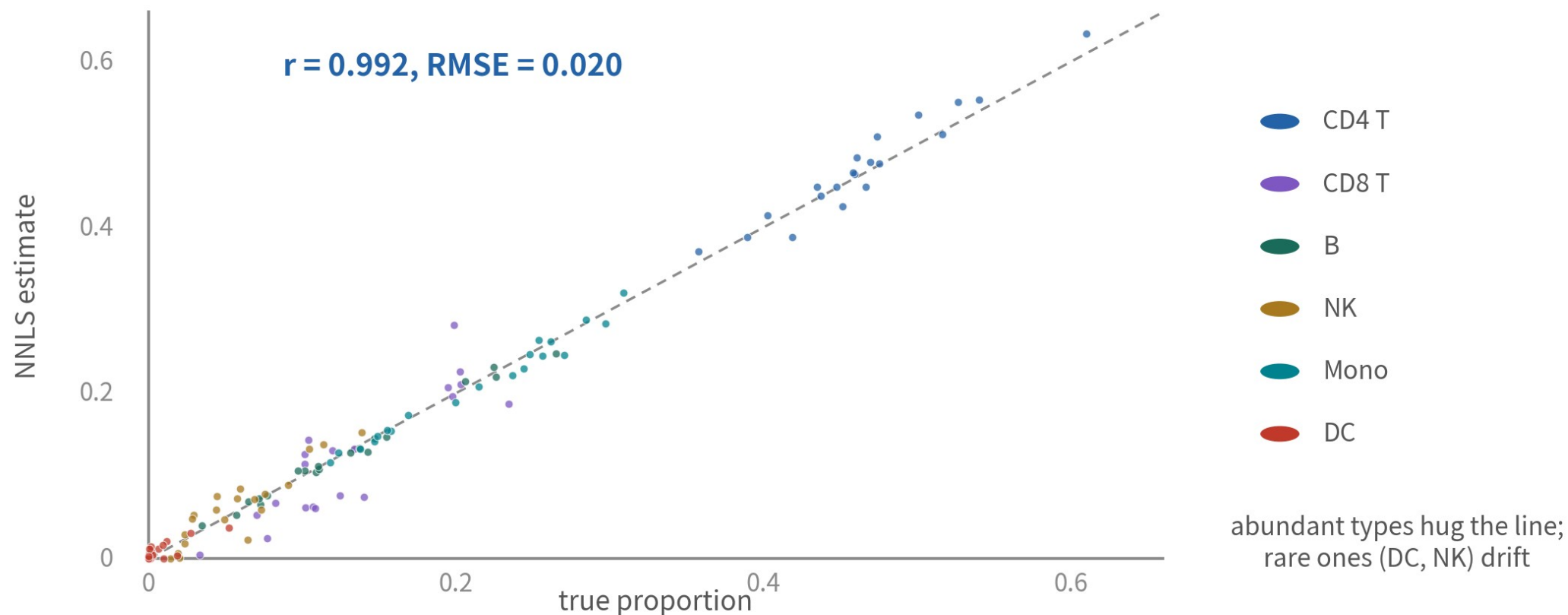
```
[1] 0.992
```

```
[1] 0.020
```

$r = 0.992$, $RMSE = 0.02$ — about two points per type on average. But overall scores lie; keep reading

Estimate vs truth: 120 points

20 pseudo-bulks $\times$ 6 types = 120 points

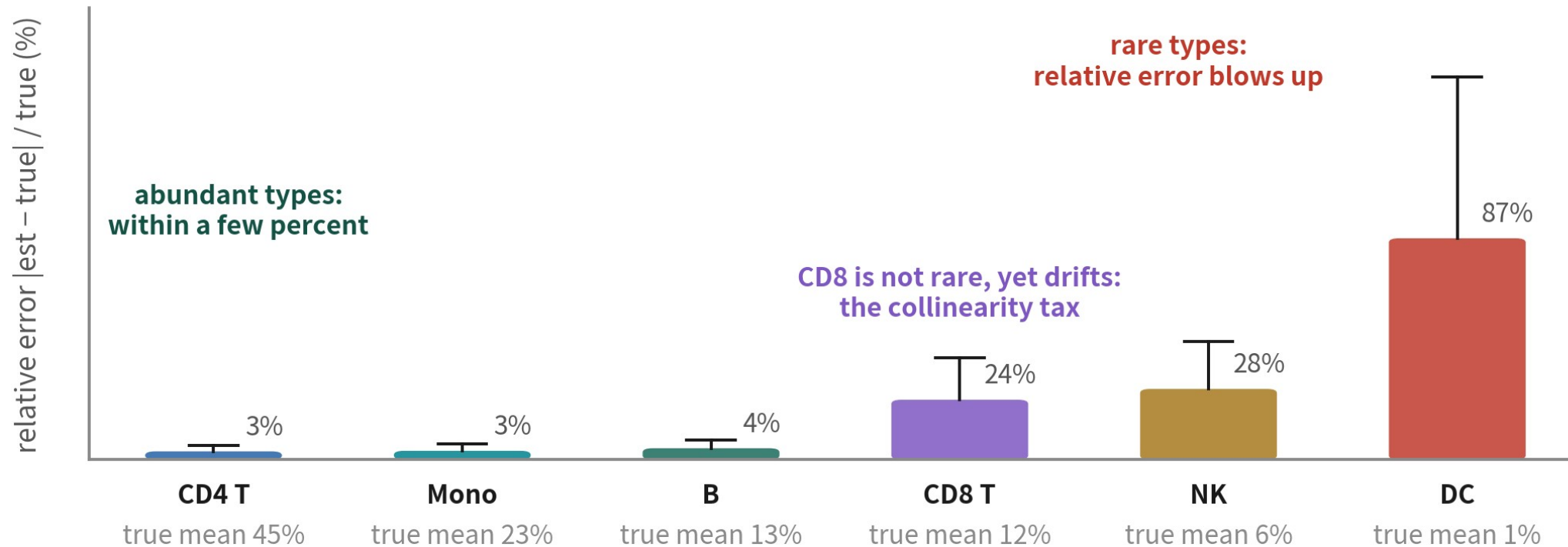


simulated data

Abundant types hug the diagonal; the rare ones crowd the lower-left and drift — zoom in on the next slide

Per-type error: abundant accurate, rare adrift

per-type recovery error (20 pseudo-bulks, real NNLS)



simulated data

CD4 / Mono / B land within a few percent; NK and DC drift 28–87% relative; CD8 drifts despite abundance — the collinearity tax

FOR YOUR METHODS SECTION

**"Accuracy was validated on simulated mixtures of known composition:
overall $r = 0.99$; rare types (<5%) showed markedly higher error."**

One sentence tells the reviewer you measured your own error — and know where it is weakest.

Three battlefields, one equation



immune infiltration in TCGA

unmix immune composition from thousands of tumour bulks, link to survival and therapy response



spatial transcriptomics spots

a Visium spot is a mini-bulk of 1–10 cells — the same equation (SPOTlight and friends)



the cross-platform reality

bulk and scRNA come from different tech; CIBERSORTx batch correction exists to bridge that gap

one equation, three battlefields — what differs is where S comes from and how noise is handled

TCGA immune infiltration, spatial spot deconvolution, cross-platform correction — all riding the equation you just solved by hand

The method family at a glance

Tool	Key idea	When to pick it
NNLS	non-negative least squares — ten lines, fully transparent	teaching, quick sanity checks, DIY pipelines
CIBERSORTx	fixed signatures + batch correction + hi-res modes	cross-platform bulk cohorts such as TCGA
MuSiC	cross-subject stability weighting on the NNLS chassis	you hold a multi-donor scRNA reference
Bisque	paired bulk + scRNA from the same cohort calibrate each other	the same subjects measured both ways
SPOTlight	NMF with signatures, aimed at spatial spots	spot-resolution spatial data like Visium

06

Where people get hurt

three traps — the first one demonstrated live

Trap 1: the reference is missing a type

What it is

A type truly present in the sample (say NK) has no column in the signature.



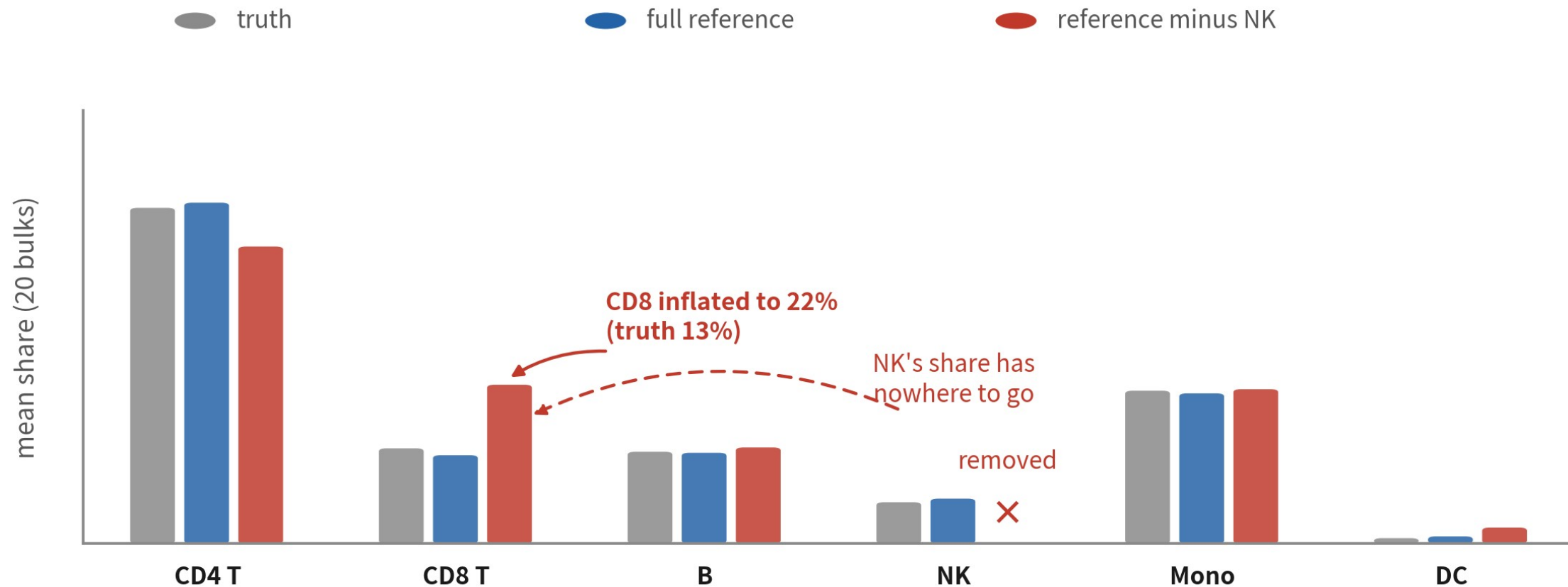
Why it happens

The solver must account for all the signal using the columns it has — the absentee's share has nowhere to go.

What it costs you

NK's share lands on its nearest kin CD8 T: in our run CD8 inflates from 12% to 22% — with no error message whatsoever.

Live run: delete NK from the reference



simulated data

Same 20 bulks, same solver, one column fewer — NK's share migrates nearly wholesale into CD8. Silent identity theft by the nearest kin

Trap 2: applying signatures across platforms, raw

What it is

Using a 10x-built (UMI, 3'-end) signature directly on full-length or microarray bulk.



Why it happens

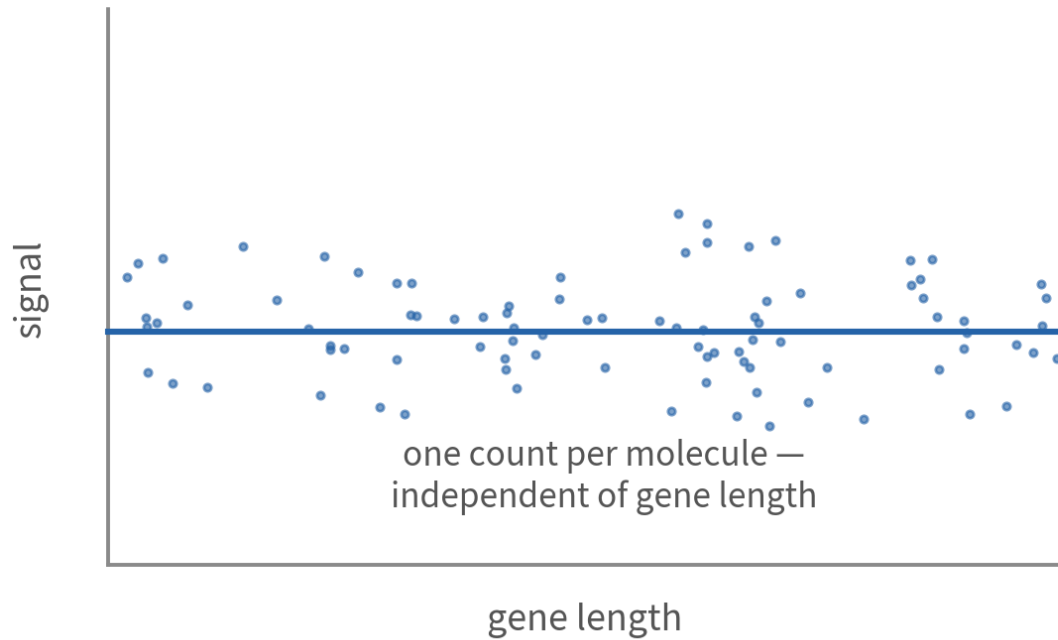
UMIs count each molecule once; full-length reads tile the transcript so long genes collect more — the scales differ.

What it costs you

Types rich in long genes are systematically inflated — the bias points one way in every sample, so a bigger cohort cannot rescue it.

Two measurement scales, one gene

10x (UMI, 3'-end counting)



full-length protocols / most bulk



one gene, two measurement scales — a 10x signature applied to full-length bulk inflates long-gene-heavy types

simulated data

This is why CIBERSORTx batch correction exists — at minimum, validate on same-platform data first

Trap 3: reading share changes as count changes

What it is

'Patient A has a higher CD8 share than B' is written up as 'A has more CD8 cells'.



Why it happens

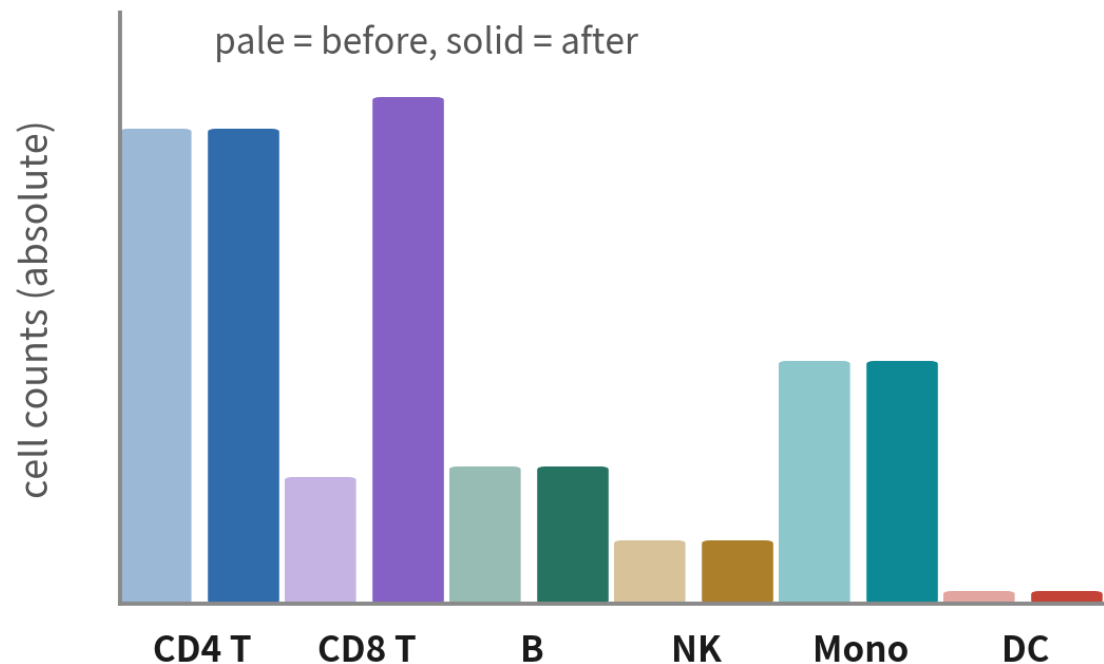
The output is compositional — locked to sum 1. When one type expands, every other share passively drops.

What it costs you

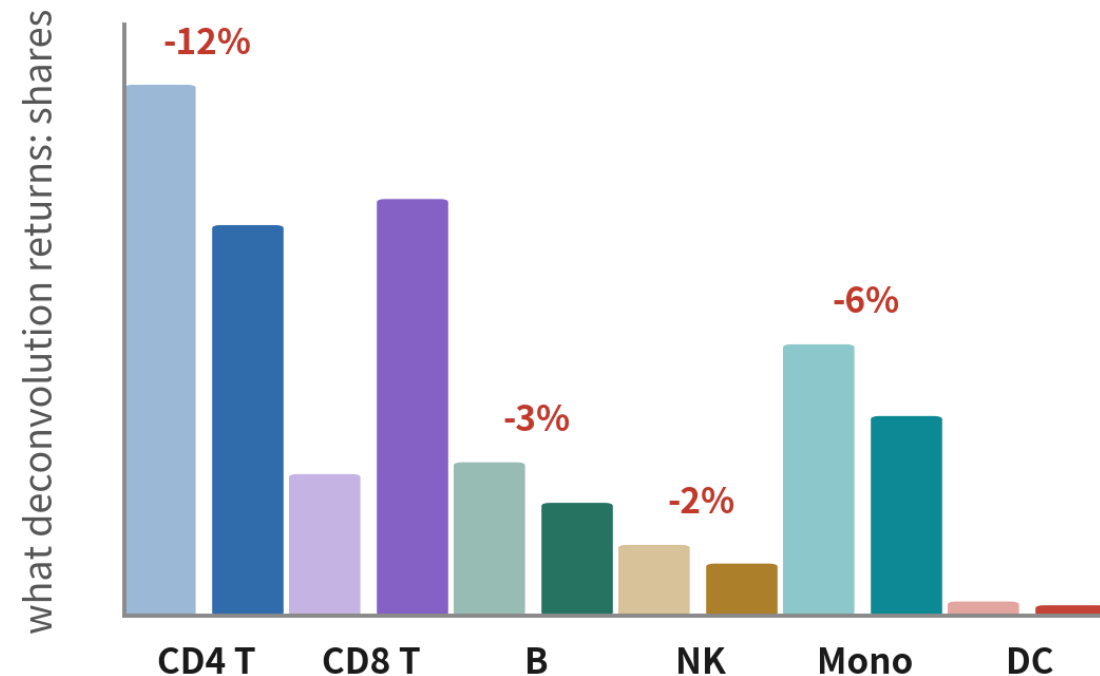
Passive drops get misread as 'B-cell depletion'; absolute claims need an external anchor, and comparisons belong on CLR-type transformed values.

Only CD8 changed; every share 'appears' to

absolute: only CD8 changed



shares: every type 'appears' to change



shares are locked to sum 1: one type expands and every other share passively drops — their counts did not fall

Left: in counts only CD8 grew. Right: in shares everything moved. Deconvolution hands you the right panel

TRAPS, SUMMED UP

**Deconvolution always returns proportions.
It never tells you the reference was wrong.**

Missing types, platform gaps, compositionality — one antidote covers all three: benchmark on pseudo-bulks with a known answer first.

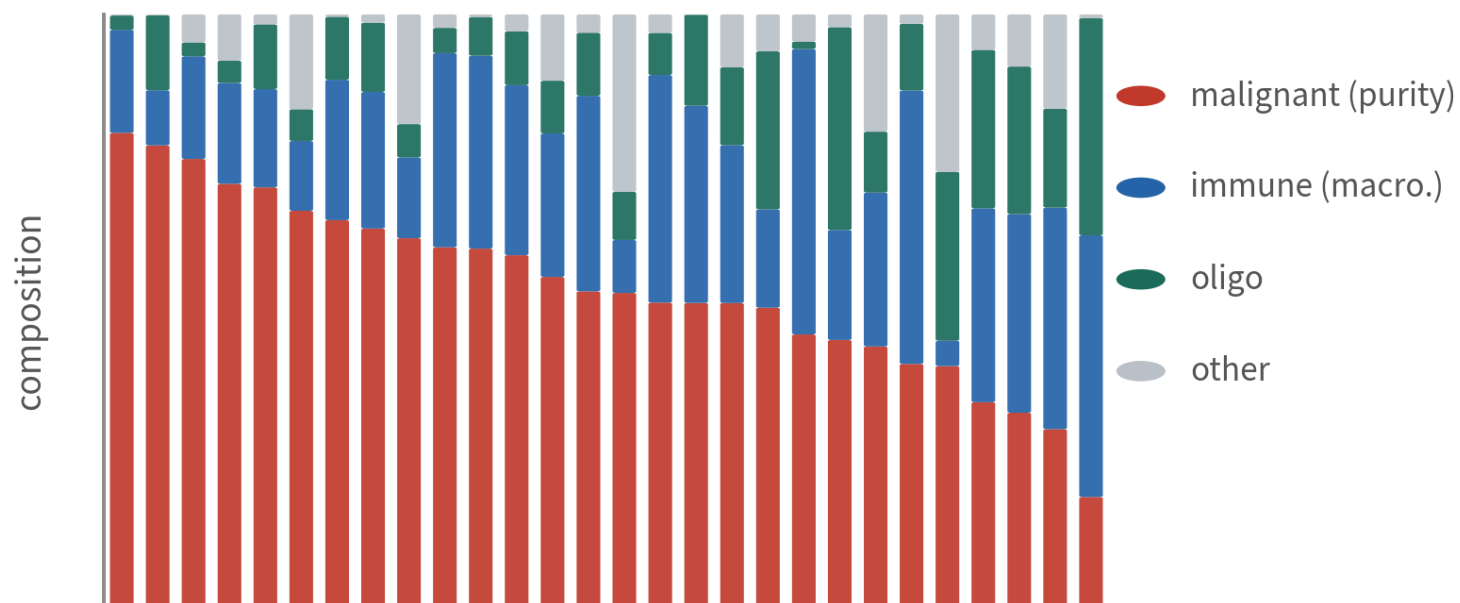
06+

Strategy Room: Complex Samples

The standard pipeline is done — now imagine feeding the equation not your own fake bulk, but real tumour bulk

Tumour bulk's first output: purity

tumour bulk composition: the malignant column is purity



each bar = one patient (sorted by malignant share)

cross-validation against published purity



you cannot mix your own ground truth for tumour bulk — but TCGA ships DNA-based purity: validate the malignant column first, then the rest
simulated data

Give the reference a malignant column and its share is tumour purity — TCGA publishes DNA-based purity estimates; validate this column before trusting any other

The malignant signature is patient-specific

Within one reference, the two kinds of columns behave nothing alike

TME types

look the same across patients

- myeloid, T, oligo: mix across patients (B9's continents)
- signatures transfer across patients and from public references
- a multi-patient reference gives MuSiC stability to weight on

Malignant cells

one island per patient, CNV-driven

- patient A's malignant signature on patient B's bulk = a mismatched reference
- the move: merge all malignant cells into one column — coarse, but it captures the shared signal
- BayesPrism goes further and lets malignant expression drift from the reference

Before proportions enter Cox: four statistical gates

Gate	Numbers from a real TCGA-GBM run	Decision
Files ≠ patients	391 files → 372 primary → 284 patients after dedup	primary only, one per patient — or deaths are double-counted
Missing follow-up	53 of 54 survivors lack follow-up; 229 analysable, 227 deaths	report between-group contrasts only, never absolute medians
Positive control	age HR 1.03 per year, $p = 1.8e-06$	the known effect shows up — only then is your null reportable
Resolution	the merged macrophage column: HR 1.10, $p = 0.46$ — null	prognosis links to specific TAM states; refine the reference and rerun first

Episode checklist



you can state the equation and name the unknown



the signature keeps discriminative markers; collinearity checked



you benchmarked on pseudo-bulks with known truth



rare-type estimates get discounted before interpretation



you asked whether the reference covers the sample's types



conclusions say 'share', never silently 'count'

All six boxes ticked before these numbers enter a paper

ONE-LINE SUMMARY

Grade yourself on mixtures you made with a known key

—

only then unmix the bulk you have no key for.

Error is not guessed at — it is measured, by you.

Check yourself 1

In $\text{bulk} \approx S \times p$, which term does deconvolution solve for?

- A The bulk expression vector
- B The signature matrix S
- C The proportion vector p
- D Gene lengths

Answer C. The bulk is measured and S comes from the single-cell reference; the only unknown is the proportion vector p . B12's pseudobulk sums forward with known labels; deconvolution splits backward with S — two directions of one identity.

Check yourself 2

What single constraint does NNLS add to least squares, and why?

- A Coefficients must sum to 1
- B Coefficients must be non-negative — a '-8% NK' is meaningless
- C There must be more genes than types
- D Residuals must be normal

Answer B. $p \geq 0$ is the one added constraint — proportions are non-negative by nature. Summing to 1 is not guaranteed by NNLS; you normalise afterwards. More genes than types is a solvability prerequisite, not NNLS-specific.

Check yourself 3

You are about to unmix tumour bulk with a PBMC reference. The biggest systematic risk?

- A Compute time
- B Insufficient bulk depth
- C The reference lacks tumour and stromal cells — their signal lands on the most similar immune types
- D NNLS errors out on tumour data

Answer C. The malignant and stromal cells dominating tumour bulk have no column in a PBMC reference, so their signal is distributed among the columns that exist — in our run, deleting just NK inflated CD8 from 12% to 22%, and here the missing types are over half the sample. The tool completes without a single warning.

NEXT EPISODE

B18: The ecosystem, and the finale

Eighteen episodes in, you can fight from FASTQ all the way to deconvolution. But this world is far bigger than R and Seurat — proteins, chromatin, space, the Python ecosystem. In the finale we unfold the whole map and point your next step.

scRNA-seq video series · Series B Hands-on